RES POLYTECHNIC, BANGALORE-91

CIE Theory Test I

Program	Computer Science and Engineering		Semester- III
Course Name	System and Network Administration		Test I
Course Code	25CS34I	Duration- 90 min	Marks-50
Name of the Course Coordinator: Vinay Kumar			
Note: Answer any one full question from each section. Each full question carries equal marks.			

Q. No	Questions	CL	CO	Marks
	SECTION I			
1	**a)** Classify the following devices/components into suitable categories (such as input, output, storage, memory, networking): **(5 marks)** i. Keyboard ii. Smartwatch Screen iii. ROM iv. Ethernet Cable v. Memory Card vi. Wi-Fi Adapter vii. Plotter viii. Webcam ix. Register **b)** Kiran is a second-year computer science diploma student. His uncle plans to set up a computer for his small business and approached Kiran for advice, assuming he knows more about computers. What hardware would you suggest for the purchase, and why? **(10 marks)** **c)** Compare the hardware components of a desktop computer with those of a smart TV. **(10 marks)**	a) L3 b) L3 c) L2	a) 1 b) 1 c) 1	25
	OR			
2	**a)** Describe the procedure to test an ATX power supply unit using a PSU tester and how the readings help you identify a fault. **(5 marks)** **b)** Divya walked across a carpeted room and touched the metal body of a CPU cabinet whose power cable was unplugged, yet she felt a sudden spark. How would you analyze the cause of this? Suggest the safety measures to be followed while handling computer hardware (static electricity). **(10 marks)** **c)** Explain the working principle of an SMPS with its stages. **(10 marks)**	a) L3 b) L3 c) L2	a) 1 b) 1 c) 1	25
	SECTION II			
3	**a)** Nisha and Arjun disagree on which chipset transfers data faster, the north bridge or the south bridge. Compare the two and conclude with a justification. **(5 marks)** **b)** While Tanvi was playing a video in a media player on a lab computer, the player stopped responding. Other files are open and unsaved, so she cannot restart the system. How can you help her close the media player? **(10 marks)** **c)** Describe the sequence of events from switching on a computer to the loading of the operating system, highlighting the role of BIOS/UEFI and POST. **(10 marks)**	a) L3 b) L3 c) L2	a) 1 b) 1 c) 1	25
	OR			
4	**a)** A lab computer gives one long and three short beeps and displays nothing on the screen. How can you use POST beep codes to locate the faulty component? **(5 marks)** **b)** The accounts office computer, in use for three years, takes a long time to boot and often freezes. Suggest the hardware and software checks and upgrades you would carry out to improve its performance. **(10 marks)** **c)** Differentiate between an SSD and an HDD on the basis of performance, price, reliability and storage capacity. **(10 marks)**	a) L3 b) L3 c) L2	a) 1 b) 1 c) 1	25

Note for the Course Coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.

Sign of the Course Coordinator Signature of the HOD Signature of the IQAC Chairman